

FIT3080 Applied Class on Introduction to Machine Learning

Exercise 1: Supervised and unsupervised learning

- (a) Identifying species of flora and fauna is more of an exercise in supervised learning or unsupervised learning?
- (b) Learning to recognize species of flora and fauna that have already been classified is more of an exercise in supervised learning or unsupervised learning?
- (c) You want to train a supervised Machine Learning algorithm for the following tasks, what will be your labelled data?
 - i. To help you predict how long will it take for you to drive home from your workplace.
 - ii. To help you recognize whether a text was written by your lecturer.

Exercise 2: Model selection and search

- (a) What is overfitting, and how can we reduce it?
- (b) Suppose we have two models that fit the training data equally well, but one is less complex. Which do we prefer for predicting new (as yet unseen) data – the simpler model or the more complex model? Which of these models is more likely to overfit the data?
- (c) Suppose we have two models that are equally complex, but one fits the training data better than the other. Which do we prefer for predicting new (as yet unseen) data?
- (d) Explain how search can be used to assist Machine Learning.

Exercise 3: Completeness, consistency and evaluation metrics

Consider the following information about two models that predict who will get a HD in FIT3080.

	Classes	
	HD	Not HD
Ground truth	A, B, C, D, E, F	G, H, I, J, K, L, M, N, O
Model A	A, B, C, D	E , F , G, H, I, J, K, L, M, N, O
Model B	A, B, C, D, E, F, G	H, I, J, K, L, M, N, O

- (a) Determine the completeness and consistency of Model A and Model B for each class.
- (b) Can a model be incomplete and inconsistent? If so, give an example.
- (c) Calculate total accuracy for Model A and Model B, and per-class recall and precision for the HD and NotHD classes for Model A and Model B.
- (d) Can you posit a relationship between completeness/consistency and recall/precision?

Exercise 4: Evaluation metrics usage: micro- and macro-averaging

Suppose you have a Machine Learning model that infers whether an elderly person has visitors in the house over 1 month (720 hours). The model has the following performance:

- Out of 80 hours when there were visitors, it detected 50.
- It also wrongly detected visitors in 20 hours when there weren't any visitors.
- It correctly detected no visitors in 620 hours.

Assuming you are measuring accuracy, which type of averaging would you use to assess the performance of this model? Why?

Exercise 5: Decision trees

In a faraway galaxy, your spaceship has just landed on an alien planet, and your crew has begun investigating the local wildlife. Unfortunately, most of your scientific equipment is broken, so all you can tell about a given object is what colour it is, how many eyes it has, and whether or not it is alive. To make matters worse, none of you are biologists (let alone exobiologists), so you intend to use a Decision Tree to classify objects near your landing site as either alive or not alive. Use the table below to answer the following questions:

Object #	Colour	# of eyes	Alive
1	Red	4	Yes
2	Black	15	No
3	Red	40	Yes
4	Green	9	Yes
5	Black	25	No
6	Red	8	Yes
7	Black	9	Yes
8	Green	15	No

- (a) What is the entropy of Alive?
- (b) What is the entropy of Alive after splitting on Colour?
- (c) What is the Information Gain of splitting on Colour?

Exercise 6: Decision trees

You work for the ATO, and you want to build a Decision Tree to predict who will cheat on their taxes. Your historical data is in this table.

ID	Tax refund	Marital status	Taxable income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- (a) Calculate the Information Gain of splitting on the categorical features **TaxRefund** and **MaritalStatus** and the continuous feature **TaxableIncome** — for **TaxableIncome**, consider splitting on the mean and the median values. Select the feature with the highest Information Gain for your first split, and draw the resultant initial Decision Tree.
- (b) Repeat the previous step for the second split, and draw the resultant Decision Tree.